5.2.3 Chemical properties of alkali metals

Alkali metals exhibit high chemical reactivity. The reactivity of alkali metals increases from Li to Cs, since the ionisation energy decreases down the group. All alkali metals are highly reactive towards the more electronegative elements such as oxygen and halogens. Some characteristic chemical properties of alkali metals are described below.

Reaction with oxygen

All the alkali metals on exposure to air or oxygen burn vigorously, forming oxides on their surface. Lithium forms only monoxide, sodium forms the monoxide and peroxide and the other elements form monoxide, peroxide, and superoxides. These oxides are basic in nature.

$4\ Li + O_2 \longrightarrow 2Li_2O$ (simple oxide)

$2\ Na + O_2 \longrightarrow Na_2O_2$ (peroxide)

$M + O_2 \longrightarrow MO_2$ (superoxide)
(M= K, Rb,Cs)

Reaction with hydrogen

All alkali metals react with hydrogen at about 673 K (lithium at 1073 K) to form the corresponding ionic hydrides. Reactivity of alkali metals with hydrogen decreases from Li to Cs.

$2M + H_2 \longrightarrow 2\ M^+H^-$
(M = Li, Na, K, Rb, Cs)

The ionic character of the hydrides increases from Li to Cs and their stability decreases. The hydrides behave as strong reducing agents and their reducing nature increases down the group.

Reaction with halogen

Alkali metals combine readily with halogens to form ionic halides MX. Reactivity of alkali metals with halogens increases down the group because of corresponding decrease in ionisation enthalpy.

$2M + X_2 \longrightarrow 2\ MX$
(M= Li, Na, K, Rb, Cs) (X= F, Cl, Br, I)

All metal halides are ionic crystals. However Lithium iodide shows covalent character, as it is the smallest cation that exerts high polarising power on the iodide anion. Additionally, the iodide ion being the largest can be polarised to a greater extent by Li^+ ion.

Reaction with liquid ammonia:

Alkali metals dissolve in liquid ammonia to give deep blue solutions that are conducting in nature. The conductivity is similar to that of pure metals (The specific conductivity of Hg is $10^4\ \Omega^{-1}$ and for sodium in liquid ammonia is $0.5 \times 10^4\ \Omega^{-1}$). This happens because the alkali metal atom readily loses its valence electron in ammonia solution. Both the cation and the electron are ammoniated to give ammoniated cation and ammoniated electron.

$M + (x + y)NH_3 \rightarrow$

$[M(NH_3)_x]^+ + [e(NH_3)_y]^-$

The blue colour of the solution is due to the ammoniated electron which

11th Chemistry Chapter-5 Vol-1 EM.indd 130 09-01-2023 10:54:34